



NASA plans to put an SMR on the moon (Source: <https://www.reddit.com>)

systems reduce the need for human oversight and use predictive diagnostics to detect issues before they escalate. Simplified HMIs offer operators real-time data to support rapid decision-making. The remote functionality and low need for manual intervention make these systems ideal for isolated locations and modular grids. Technological advancements continue to enhance their reliability and adaptability.

Human machine interface (HMI) systems. HMI systems provide intuitive, adaptable interfaces for seamless navigation and efficient onboarding in high-pressure environments. In SMRs, HMIs enable reliable plant operations and provide real-time visibility of reactor performance, facilitating informed decision-making.

Operators use HMI displays to monitor thermal output, coolant flow, and grid synchronisation. These systems facilitate continuous data exchange, enforce control logic, and support alarm management for timely interventions. Remote monitoring capabilities offer flexibility,

particularly in distributed or remote deployments.

Modern HMI systems are evolving to support cloud-based connectivity, integrate augmented and virtual reality, and harness advanced analytics for predictive maintenance, diagnostics,

and operator training.

Cybersecurity integration.

Cybersecurity protects systems, networks and programmes from cyber-attacks, typically aimed at accessing, altering or destroying sensitive information, extorting money, or interrupting normal operational processes. Digital I&C systems are vulnerable to cyber threats targeting control logic, sensor data integrity, and command-and-control communications. To mitigate such risks, SMR designs incorporate robust security protocols, logical isolation between safety and non-safety systems, and multi-factor authentication. Indigenous designs embed cybersecurity at the system architecture level, rather than as an add-on layer, to ensure resilience against cyber intrusions from the outset.

Emerging and advanced technologies. Emerging technologies such as artificial intelligence (AI), machine learning (ML), field-programmable gate arrays (FPGAs), advanced sensor networks, digital twins, and high-performance data processing systems are increasingly

Box 3: Field programmable gate arrays (FPGAs)

FPGAs are configurable integrated circuits that can be repeatedly programmed after manufacturing. They are a subset of programmable logic devices, used for deterministic execution, hardware-level reliability, and resistance to common software failures. Designers see significant advantages in FPGA-based I&C systems over microprocessor-based systems, as these are simpler, more testable, less reliant on complex software, and easier to qualify for safety applications.

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